

NDNSF-UAV: Map-Selected Multi-UAV Inspection

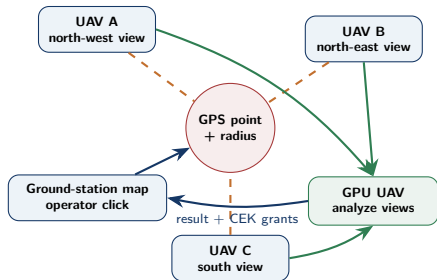
Late-bound camera UAVs capture complementary views; a selected GPU UAV analyzes them and returns named results

WORK IN PROGRESS

DATA-DRIVEN

REALISTIC MVP

Concrete Case: Capture, Analyze, and Display One GPS Point



One concrete task

- ▶ The operator clicks a location; the planner selects three camera roles and one GPU-capable Analyzer.
- ▶ Each camera publishes one encrypted ViewImage; the Analyzer fetches admitted objects by name and processes them.
- ▶ The Analyzer publishes per-image annotations and requester-wrapped CEK grants. The Requester fetches the same image objects and overlays the annotations locally.

Why this is a realistic first case

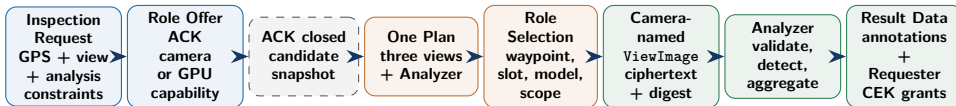
The known location removes target association and triangulation; the MVP reuses an existing per-view detector.

Key point

Each immutable image object serves both Analyzer and Requester; only compact annotations and recipient-bound CEK grants are newly published.

Evidence: Multi-UAV visual inspection and coverage systems already distribute image-acquisition work across aircraft [1,2].

Fully Data-Driven NDNSF Workflow



Request and planning

- ▶ Request: point/radius, time window, view diversity, image quality, model requirement, and result deadline.
- ▶ Offer: cameras report reachability and view capability; processors report GPU, model, queue state, and input limits.
- ▶ Plan: View-A/B/C plus one Analyzer; named edges connect each View role to the Analyzer.

Execution and result admission

- ▶ Selection assigns viewpoints and binds the Analyzer, Requester, named dependencies, model, and deadline.
- ▶ Each Camera publishes once under `/<camera>/NDNSF/COLLAB/...`, with request/plan/attempt, capture metadata, and a CEK wrapped to the Analyzer.
- ▶ The Analyzer requires prefix, signature, Plan role-Provider, and dependency-edge agreement. Its own signed result names every image/digest and carries annotations, failures, and the same CEKs rewrapped to the authorized Requester.

ACK SNAPSHOT

ONE PLAN

REUSED VIEW DATA

RESULT + KEY GRANT

EXPLICIT FAILURES

The Requester fetches original images from cache or Repo, decrypts them, and overlays the returned annotations.

Evidence: Current API: ACK closure and plan commit. In progress: cross-role input/result enforcement.

Practical Boundary, MVP, and Required Evidence

Responsibility boundary

- ▶ **NDNSF**: protected Request, discovery, ACK closure, producer-owned names, signer/role/dependency admission, recipient-bound CEK grants, and failure state
- ▶ **UAV application**: viewpoint/navigation, capture metadata, analysis invocation, annotation schema, and Requester-side overlay
- ▶ **Perception backend**: existing per-view detector and deterministic evidence aggregation; it does not decide Provider assignment
- ▶ **Autopilot**: geofence, collision, readiness, and final safety rejection

Current implementation work. Complete dependency/result admission and Requester-bound CEK grants; integrate frame metadata and annotations.

End-to-end claim: each encrypted image is published once, analyzed remotely, and securely reused by the Requester.

Evidence: [1] Mansouri et al., Control Engineering Practice 2018; [2] Luna et al., Journal of Field Robotics 2024; [3] MAVLink Common Message Set.

First measurable MVP

- ▶ one static outdoor GPS point, three camera roles, three camera UAVs, one GPU Analyzer, and one existing detector
- ▶ test provider unavailability, mobility, Data loss, missed deadlines, and unsafe-assignment rejection

Measurements and ablations

- ▶ completion, valid-view and analysis success, latency, bytes, image publications per capture, and flight/energy proxy
- ▶ one versus three views under occlusion; nearest versus geometry-aware selection; Analyzer failure and retry

Claim boundary: no new recognition algorithm, moving-target identity, fleet autonomy, triangulation, or 3-D accuracy claim.